

switching to irradiated-sterilized acetate-membrane or wet-membrane dialyzers

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented nephrology problem: maintenance hemodialysis patients receive a prescribed dialysis dose (commonly indexed as single-pool Kt/V, with guideline targets reported around 1.2-1.4 per session in the literature) that depends directly on the sustained blood flow rate through the extracorporeal circuit and the dialysate flow rate across the dialyser. When the delivered blood flow rate falls below the prescribed value (prescriptions commonly in the 300-500 mL/min range in the literature, with some reports of inadequate dose at lower effective flows), the delivered dose falls short and session time must be extended. Documented contributing factors include vascular access recirculation, arterial and venous pressure limits of the circuit, dialyser mass-transfer area and fiber bundle geometry, dialysate flow proportioning limits, and needle gauge resistance. This description is qualitative problem-domain context from the nephrology and biomedical-engineering literature; no specific incidence rates are asserted here (the engine's own retrieval must find and custody the evidence).

Mechanism: hypersensitivity to the hemodialyzer membrane (n = 11, 27.5%)

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: switching to irradiated-sterilized acetate-membrane or wet-membrane dialyzers

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * dP) / (8 * \mu * L)$. Applicability to this invention: CONDITIONAL; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: NOT ESTABLISHED.

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$. Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.